Transitioning to New PDB IDs and PDBx/mmCIF Format at the Protein Data Bank

wwpdb.org

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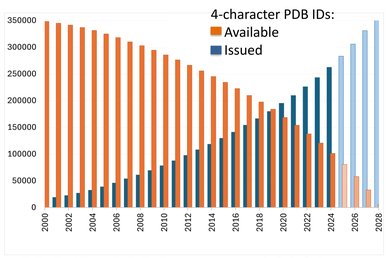
Introduction

Due to rapid growth in structure depositions, the wwPDB anticipates exhausting the current 4-character PDB IDs by 2028. To prepare for this, the PDB is transitioning to extended IDs, formatted as "pdb_" followed by eight alphanumeric characters (pdb_xxxxxxxx). New entries will be available **only** in PDBx/mmCIF format.



PDB ID: 1ABC -> pdb_00001abc

Challenges:

- Since 1971, over 230,000 entries have used 4-character PDB IDs
- Software frequently built around legacy file format that cannot support extended IDs
- Users, journals, and software developers need to adopt new PDB IDs and PDBx/mmCIF format now

Project goals

The PDB archive supports 60,000 Data Depositors and millions of Data Consumers who use the data across a variety of scientific disciplines.

By 2028, all PDB users will need to adapt to using *only*

- **Extended PDB IDs**
- **The modern file format (PDBx/mmCIF)** used by the PDB archive for data deposition, curation, validation, and distribution, replacing the legacy format that cannot support extended IDs
- **A new directory structure** for the PDB archive that supports the new PDB IDs

Key milestones: 5-year transition plan

A detailed plan was developed for the gradual implementation of extended PDB IDs, with full adoption targeted for 2028.

2023

- **Provide training materials for adoption of PDBx/mmCIF and extended PDB IDs**

2024

- **Create PDBx/mmCIF User Guide**
- **Update archival PDBx/mmCIF files with extended PDB IDs**

2025

- **Provide PDBx/mmCIF example files at GitHub**
- **Register new PDB DOIs based on extended PDB IDs**

2026

- **Establish "beta" PDB archive designed around extended PDB IDs with new file directory architecture**

2027

- **Support extended PDB IDs by OneDep deposition system and wwPDB partner websites**
- **Release "beta" PDB archive as PDB main archive**

Available transition resources

Dedicated webpage describing the project

Documentation, FAQs, software tools, etc.
https://www.wwpdb.org/documentation/new-format-for-pdb-ids

Extended IDs in the public files – ready for testing

For the current entries, the extended PDB IDs are generated by adding prefix "pdb_0000" to their 4-character IDs:

4HHB / pdb_00004hhb

All current archival files have been updated with the extended PDB ids, users can start testing /adopting the extended IDs now:

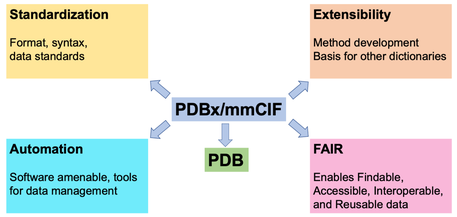
_database_2.database_id	*PDB*
_database_2.database_code	*4HHB*
_database_2.pdbx_database_accession	*pdb_00004hhb*

Example files at GitHub – ready for testing

https://github.com/wwPDB/extended-wwPDB-identifier-examples

PDBx/mmCIF data format and resources

PDBx/mmCIF – PDB master archival format since 2014

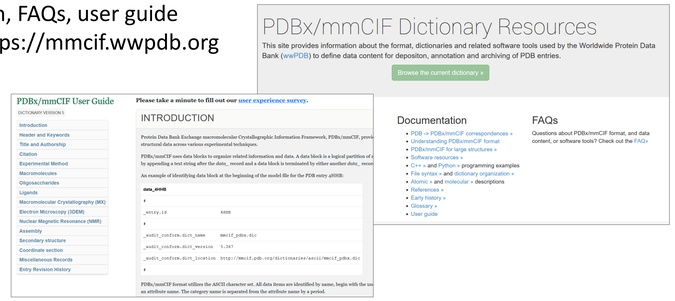


- Community-developed
- Supports structures of any size and complexity
- Defines data types, boundary conditions, enumerations

Westbrook, et al., (2022) JMB 434: 167599

Entries with extended PDB IDs will be available only in PDBx/mmCIF format.

Documentation, FAQs, user guide available at https://mmcif.wwpdb.org



Major structural biology software tools support PDBx/mmCIF

- Software libraries, parsers, format converters, file editors
- Visualization tools: Chimera, Mol*, PyMol, Jmol, Coot, CCP4mg
- Refinement programs: Phenix, Refmac, Buster

Actions required from community

Data depositors

Data depositors should report the extended PDB IDs in their publications

Data users

Users need to adopt PDBx/mmCIF format now
Users should adopt to new archival file organization and file naming:

All data files for an entry will be stored in a single directory named by the **penultimate two characters** of the extended PDB code.

For pdb_00001abc: *pub/pdb/data/entries/ab/pdb_00001abc/*

To aid users in adapting to this change, a "beta" archive will be provided during the transition phase. Testing starts in 2026.

Journal editors

Journal editors should adopt extended PDB IDs and associated DOIs in their pipelines:

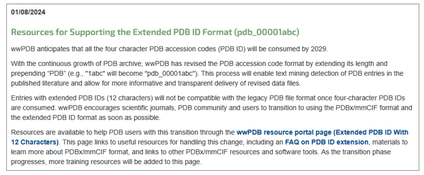
pdb_00001abc | 10.2210/pdb_00001abc/pdb

Software developers

Software developers should adjust their tools to work on PDBx/mmCIF formatted files

Our communication strategies

- News items
- Community mailing lists (CCP4, CCN, ccpem, 3DEM, EM-UK, DROR, CCPN, BMRB, pdb-l)
- Social media
- Virtual office hours, webinars
- Flyers
- Professional society meeting presentations
- Email communication with scientific journals and community software developers



pdb_ 00001 abc

Register for the March 13 Virtual Office Hour on Supporting Extended PDB IDs

Interested in learning more about plans to extend PDB IDs to 12 characters? Bring you questions to a virtual Office Hour.

Feedback needed!

We are seeking feedback to ensure a smooth transition to new PDB IDs and PDBx/mmCIF file format.

- Are we reaching the community effectively?
- Are the right channels being used?
- Are our materials and tools sufficient?
- Any additional resources needed?

Questions? Comments? Contact us at: deposit-help@mail.wwpdb.org

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